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RESEARCH STATEMENT

Spectral analysis and adaptive filtering for visual perception, from recognition to image generation. My research is about visual perception for machine intelligence: I analyze what vision models pass or suppress in the spectral domain and turn that analysis into operators for visual recognition and image generation.

- Spectral analysis of convolution and self-attention (frequency balancing, graph-spectral frequency responses)
- Spectral-adaptive operators as general-purpose replacements for softmax attention, from vision backbones to diffusion transformers (SPAM, SALA)
- Visual perception for robots and healthcare (fall detection, isolation-ward services, daily-life monitoring)

In the next step, I plan to do the same for video generation and world models at a larger scale.

EDUCATION

Feb 2027 (expected) **Ph.D. in Computer Science and Engineering**
Korea University Seoul, Korea
Dissertation: *Spectral Analysis and Adaptive Filtering for Visual Perception*

2018 **M.S. in Mechatronics**
Gwangju Institute of Science and Technology (GIST) Gwangju, Korea
Thesis: *Removal of Jitter Noise using a Modified Kalman Filter in 3D Shape Recovery from Focus*

2016 **B.S. in Control and Measurement Engineering**
Gyeongsang National University (GNU) Jinju, Korea

RESEARCH EXPERIENCE

Since 2019 **Student Researcher**
Intelligence and Interaction Research Center, **Korea Institute of Science and Technology (KIST)**
Supervisor: Dr. Dong Hwan Kim

Linear Attention for Generative and Vision Transformers (SALAFormer; in preparation [11])

- Designed **SALA** (Spectral-Adaptive Linear Attention) as closed-form associative memory: each head solves a small ridge regression for what to store in its fixed-size memory, instead of summing raw key–value products, and a learned spectral filter on the key Gram matrix allocates that memory, at linear $O(N)$ inference cost.
- **Image generation:** lower FID than MHLA (ICLR 2026), a recent linear-attention design for diffusion transformers, at both model sizes: 59.80 → **57.65** (DiT-S/2) and 37.47 → **36.54** (DiT-B/2) vs. its published results (class-conditional ImageNet 256×256, no guidance).
- **Image classification:** SALA in a vision backbone (SALAFormer) reaches **77.2%**

/ **81.8%** ImageNet-1K top-1 at DeiT-Tiny/Small size, above the 76.5% / 81.6% reported for ViT^3 (CVPR 2026) with the same recipe.

Spectral Analysis of Convolution and Attention for Vision Backbones

- Developed **SPANetV2**: compared what convolution and self-attention pass or suppress in one graph-spectral framework, showing that window size shapes their frequency responses, and turned this into a mixer with multi-scale kernels and spectral re-scaling; the backbone outperforms state-of-the-art models on ImageNet-1K, COCO, and ADE20K. [1]
- Proposed **SPANet**: prior work found that self-attention acts as a low-pass filter, while convolution retains high frequencies, and that stronger high-pass filtering helps self-attention. SPANet showed that stronger low-pass filtering also helps convolution, and that a token mixer *balancing* high and low frequencies improves accuracy. [4]

Applied Vision for Robotics, Healthcare & 3D Shape Recovery

- Fall detection of older adults by a healthcare robot [6]; robot-assisted services in isolation wards [5]; daily-life arousal monitoring and brain–computer interface decoding with egocentric vision and EEG [2, 3]; a view-consistent evaluation metric for multi-view panoptic segmentation [10].
- Shape from focus, a follow-up to the M.S. thesis: 3D shape from few images with Gaussian-process regression [7].

2018 – 2019 **Intern Researcher**
Center for Intelligent Robotics, **KIST**

- Perception for healthcare robots: detecting when people with dementia leave home, to keep them from going missing [9]; accelerometer–depth fall detection for older adults [8], later extended with LSTMs [6].

PUBLICATIONS

Published / Accepted

[1] Guhnoo Yun, J. Yoo, K. Kim, J. Lee, P.H. Seo, D.H. Kim, **Spectral-Adaptive Modulation Networks for Visual Perception.** *IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI)* 48(10), 11511–11525 (2026). [doi: 10.1109/TPAMI.2026.3690455] [arXiv] [code]

[2] W. Hyung, M. Kim, Y.-S. Kim, J. Lee, B. Ko, Guhnoo Yun, D.H. Kim, C.-H. Im, **Context-Aware Daily-Life Arousal Monitoring Using Egocentric Vision and EEG.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)* (2026). [program]

[3] Y.-S. Kim, M. Kim, W. Hyung, B. Ko, Guhnoo Yun, D.H. Kim, C.-H. Im, **PeBCI: Vision-Guided Brain-Computer Interface for Context-Aware Neural Decoding in Daily Life.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)* (2026). [program]

[4] Guhnoo Yun, J. Yoo, K. Kim, J. Lee, D.H. Kim, **SPANet: Frequency-balancing Token Mixer using Spectral Pooling Aggregation Modulation.** *IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 6090–6101 (2023). [doi: 10.1109/ICCV51070.2023.00562] [CVF] [arXiv] [project] [code]

[5] Y. Kwon, S. Shin, K. Yang, S. Park, S. Shin, H. Jeon, K. Kim, Guhnoo Yun, …, Y. Lim, **Heterogeneous Robot-Assisted Services in Isolation Wards: A System Development and Usability Study.** *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 8069–8076 (2023). [doi: 10.1109/IROS55552.2023.10341857]

[6] K. Kim, Guhnoo Yun, S.-K. Park, D.H. Kim, **Efficient Fall Detection for a Healthcare Robot System Based on 3-Axis Accelerometer and Depth Sensor Fusion with LSTM Networks.** *Int. Conf. on Pattern Recognition (ICPR)*, 2207–2212 (2022). [doi: 10.1109/ICPR56361.2022.9956418]

[7] H.-S. Jang, Guhnoo Yun, M.T. Mahmood, M.-K. Kang, **Optimal Sampling for Shape from Focus by Using Gaussian Process Regression.** *IEEE Int. Conf. on Consumer Electronics (ICCE)*, 1–4 (2020). [doi: 10.1109/ICCE46568.2020.9043150]

[8] K. Kim, Guhnoo Yun, S.-K. Park, D.H. Kim, **Fall Detection for the Elderly Based on 3-Axis Accelerometer and Depth Sensor Fusion with Random Forest Classifier.** *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 4611–4614 (2019). [doi: 10.1109/EMBC.2019.8856698]

[9] Guhnoo Yun, K. Kim, S.-K. Park, D.H. Kim, **A Monitoring System to Support Home Health Care for the Elderly with Dementia by Detecting Going Out Activities Based on RGB-D Sensors.** *Int. Conf. on Ubiquitous Robots (UR)*, 71–76 (2019). [doi: 10.1109/URAI.2019.8768598]

Under Review

[10] Y. Lee, B. Ko, Guhnoo Yun, D.H. Kim, **Every View Counts: View-Consistent Panoptic Quality for Multi-view Panoptic Segmentation.** *arXiv preprint* (2026). [arXiv: 2610.05911]

In Preparation

[11] Guhnoo Yun et al., **SALAFormer: Spectral-Adaptive Linear Attention as Closed-Form Associative Memory.**

PATENTS

2025 **Frequency-Balancing Token Mixer and Its Backbone Model**
Korean Patent No. 10-2815184 (registered May 27, 2025)

2022 **System and Method for Determining Whether a Resident Has Left Home**
Korean Patent No. 10-2374057 (registered Mar 8, 2022)

AWARDS & SCHOLARSHIPS

2026 **Outstanding Student Researcher Scholarship**
Korea Institute of Science and Technology (KIST)

2023 **Outstanding Technology Award**
Korea Institute of Science and Technology (KIST)

2020 **Presidential Award** 2020 International Robot Contest—Humanoid Robot Sports (Deep Learning Algorithm)
Ministry of the Interior and Safety

2016 – 2017 **Graduate Scholarship**
Gwangju Institute of Science and Technology (GIST)

2013 – 2015 **Academic Excellence Scholarship**
Gyeongsang National University (GNU)

INVITED TALKS

2026 **2026 Inter-Ministerial Public Technology Transfer and Commercialization Roadshow**
Presentation at the Science and Technology Convention Center, Seoul

2021 **"Recent Research Trends in Computer Vision"**
Seminar at Korea Polytechnic University (now Tech University of Korea)

2021 **"6D Object Pose Estimation"**
Seminar series at CUBOX (now SECERN AI)

PROFESSIONAL SERVICE

2024, 2026 **Reviewer, International Conference on Pattern Recognition (ICPR)**

TECHNICAL SKILLS

Languages & Frameworks: Python, PyTorch, C++, MATLAB; timm, MMDetection, MMSegmentation, ROS

REFERENCES

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